

Problem Set 4

A. Basketball Exercise

MAIN

ACMSGURU

PROBLEMS

SUBMIT

STATUS

STANDINGS

CUSTOM TEST

☒ my only
 ☐ friends only

Contest status

#	When	Who	Problem	Lang	Verdict	Time	Memory
379799198	Jun/22/2026 22:10 ^{UTC+10}	tianzheyu	580D - Kefa and Dishes	C++23 (GCC 14-64, msys2)	Accepted	328 ms	44900 KB

<https://codeforces.com/submissions/tianzheyu#>

Process

There are n dishes each with a satisfaction level. And there are k rules each has 3 values x, y, z such that if the person takes dish x and immediatly takes dish y , he can get extra z satisfaction. The task is to find the maximum overall satisfaction he can get after taking m dishes.

Challenges and Overcoming

Since $1 \leq m \leq n \leq 18$, this is a hint from using bitmask dp. Bitmask can be used as a set to keep track of which dishes we have already taken. And since we can get extra satisfaction if we take 2 dishes continuously as in the rules, we also need to keep track of the previous dish we've already taken. Therefore, our dp table looks like this:

Rows: Chosen Dishes Set mask (binary)		Columns: The Last Dish Eaten (last_dish)			
		Dish 0	Dish 1	Dish 2	
001 (Only dish 0)		sat[0]	-1	-1	
010 (Only dish 1)		-1	sat[1]	-1	
011 (Dishes 0 & 1)		dp[011][0]	dp[011][1]	-1	
100 (Only dish 2)		-1	-1	sat[2]	
101 (Dishes 0 & 2)		dp[101][0]	-1	dp[101][2]	
110 (Dishes 1 & 2)		-1	dp[110][1]	dp[110][2]	
111 (Dishes 0, 1 & 2)		dp[111][0]	dp[111][1]	dp[111][2]	
					

Base Cases

The base cases should be if we only take 1 dish, there is no previously taken dish, so the satisfaction natually be the satisfaction of that dish only.

Transform Function

For each cell (mask, i), we will first check if i is in the bitset, otherwise it makes no sence that we have perviously taken the dish i if i is not even in the set of dishes have been taken. Then, we will get the satisfaction from the previous stage (i.e. with no i in the set (`mask & ^ (1 << i)`)), we will loop though that

row and get the maximum satisfaction in that stage. And in this process, we will check if there is a rule `rule[q][i]` (satisfaction we get if first take dish q and then take dish i). We will get the maximum and fill that into the cell.

Finial Answer

To get the finial answer, we just loop though the table, and for each bitset with m elements, we keep track of the largest satisfaction we get. And that is the finial answer.